English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Computer Science, 7 June 2025 Test 2 ——— *The exact grading formula depends on the results. As an approximation, use 2 + the number of points. Time allowed: 3 hours. Good luck!*

Problem 1. (0.5 points) A printer costing 100 BGN has an exponentially distributed lifetime with mean 3 years. Under a two-year warranty, the retailer refunds the entire price if it fails in the first year and half if it fails in the second. Let S_{100} be the total refund for 100 sold printers. Find the expectation and variance of X.

Problem 2. Two instruments measure a building's height h. Both errors are normal with mean 0; their standard deviations are $0.04h$ and $0.05h$. Measurements are independent.

1. (0.25 points) For each instrument, find $\mathbb{P}(|\text{error}| < 0.03h)$.
2. (0.25 points) What is the probability that the average of the two measurements differs from h by no more than $0.03h$?
3. (0.25 points) With one measurement from each instrument, which linear combination has mean h and the smallest standard deviation?
4. (0.5 points) Using only instrument 1, approximately how many measurements are needed for error below $0.01h$ with probability at least 95%?

Problem 3. Let (X, Y) have joint density $f_{X,Y}(x, y) = cx/y^2$ for $0 < x < y < 1$, and 0 otherwise.

1. (0.75 points) Find c, $\mathbb{E}[X]$ and $\mathbb{E}[X \mid Y = 1/3]$.
2. (0.25 points) Find the distribution of $Z = X/Y$.

Problem 4. Let $U \sim U(0, 1)$ and $V \sim U(-\pi/2, \pi/2)$.

1. (0.5 points) A method of generating a pseudorandom number between 0 and 1 from U is to choose a typically large positive integer M and take the fractional part $X = \{MU\}$. Find $\mathbb{E}X$, DX and $\text{Cor}(U, X)$. What do you think of this method?
2. (0.25 points) Let $U_1, \ldots, U_{100}$ be iid copies of U. What is the expectation of each, and of their maximum and minimum?
3. (0.5 points) Find the density, expectation and variance of $Y = \tan V$.
4. (0.25 points) Let $Z \sim \text{Cauchy}(1)$, with density $1/[\pi(1 + x^2)]$ on $\mathbb{R}$. Prove that $1/Z$, $2Z/(1 - Z^2)$ and $(3Z - Z^3)/(1 - 3Z^2)$ have the same distribution.

Problem 5. *Bonus* (0.25 points) Find $\lim\limits_{n\to\infty} e^{-n}\left(\dfrac{n}{1!} + \dfrac{n^2}{2!} + \cdots + \dfrac{n^n}{n!}\right)$.